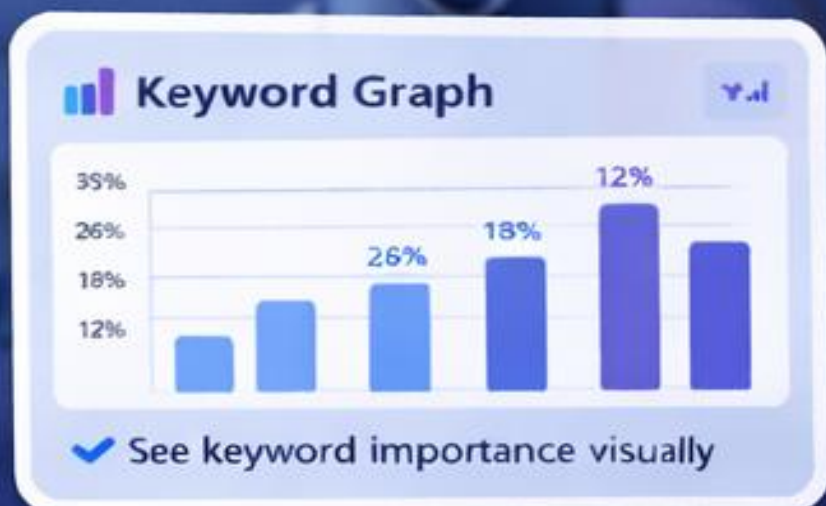




MeetText

Automated Meeting Summarization System

"Transforming speech into searchable insights."

-  Transcribes meeting audio
-  Segments discussion topics
-  Summarizes key points

Built for better memory, clarity & productivity

KOWSALYA S

Problem statement and solution



Meetings often result in long and unstructured conversations that are difficult to remember, summarize, or share with others. Important points can easily be forgotten or missed, especially when there are no clear notes or summaries available. This project aims to solve that problem by creating a system that automatically listens to meeting audio, breaks it down into topics, and provides a clear summary of the discussion. This helps save time, improves understanding, and makes it easier to revisit key information whenever needed.



Project Objectives



**Convert
spoken
meeting
content into
written text**

**Identify and
separate
different
discussion
topics**



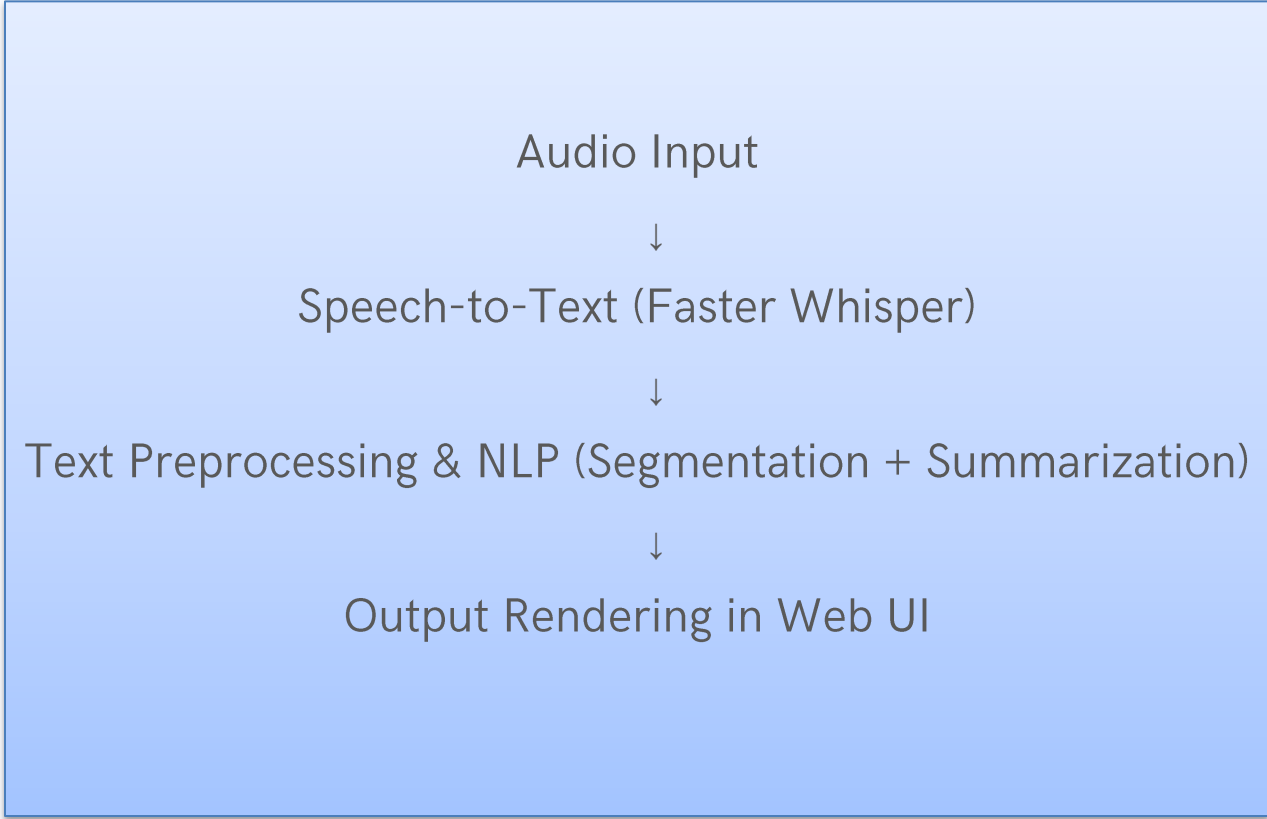
**Generate concise
summaries and
extracts key insights**

**Provide a user-
friendly interface
for navigating the
contents**



**Support
knowledge
sharing by
making meeting
content easier
to revisit and
share with
others.**

System Architecture



The system processes audio locally, extracts speech as text, segments topics using NLP techniques then summarises and visualizes the result in a minimal web interface.



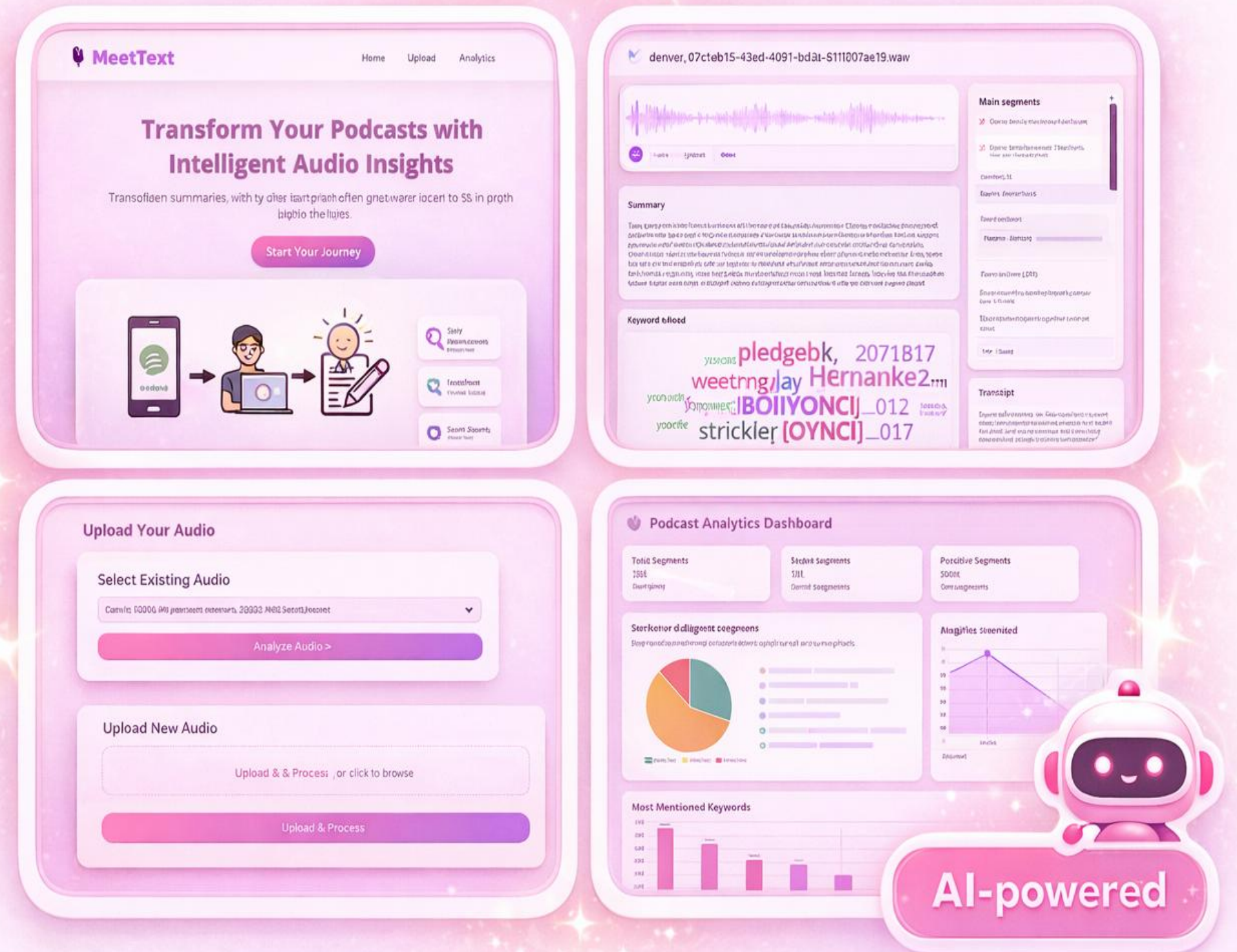
Technologies



USAGE of UI

The user interface (UI) for *MeetText* was designed to ensure a smooth, intuitive experience while interacting with complex audio analysis. It includes:

- A **homepage** to explain the system's purpose and invite users to start processing their audio.
- An **upload module** for selecting or dragging-and-dropping audio files easily.
- A **summary and transcription screen**, displaying generated text, keyword cloud, and segmented timestamps.
- An **analytics dashboard** with keyword graphs and sentiment insights to make meeting content more actionable.



*The UI was styled in a soft **pink-purple theme** to reflect a calm, modern workspace, combining functionality with approachability. It was created using clean layouts, responsive components, and animated visuals to enhance user engagement*



LIMITATIONS AND CHALLENGES FACED

1.Accuracy with Noisy Audio

Background noise and overlapping speakers reduced transcription quality.

2.Segmentation Sensitivity

Topic segmentation sometimes split unrelated or very short segments due to inconsistent pacing or vague context.

3.Performance Bottlenecks

Processing large files took longer and occasionally caused UI delays without optimization.

4.Summarization Balance

Creating summaries that were neither too vague nor too detailed was difficult without extensive model fine-tuning.

5.Frontend + Backend Sync

Aligning the UI flow with backend processing stages (especially status updates) required careful asynchronous handling

Future scope

- Speaker diarization (who said what)
- Multilingual transcription support
- Real-time processing
- Integration with Zoom, MS Teams, or Google Meet



Looking Ahead with MeetText

- Every word in a meeting matters— MeetText helps capture it.
- We've built more than transcription: a way to turn speech into insight.
- With clarity, simplicity, and care— MeetText is ready to grow.

